

Geometry

Unit 12

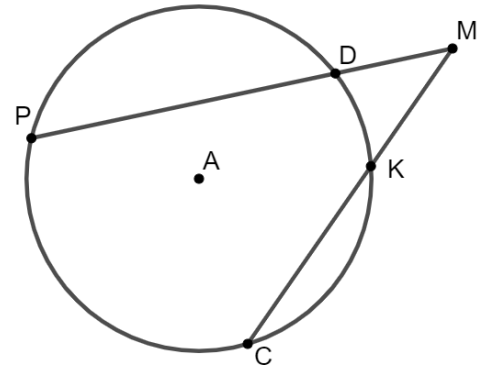
Lesson 2 Homework

Name _____

Date _____

Period _____

Circle A with secants $\overline{PM}$ and $\overline{CM}$ is shown. $\overline{PM}$ and $\overline{CM}$ intersect at point M outside the circle so that $PD = x$, $DM = 3$, $CK = 8$, and $KM = 4$.

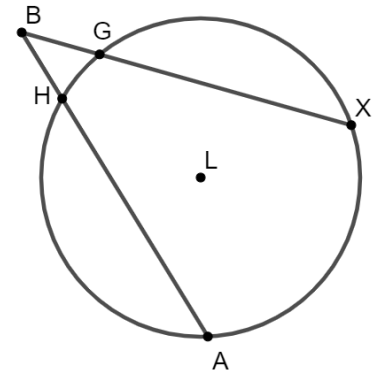


1. What is the length of secant $\overline{CM}$?

2. Find the value of x .

3. What is the length of secant $\overline{PM}$?

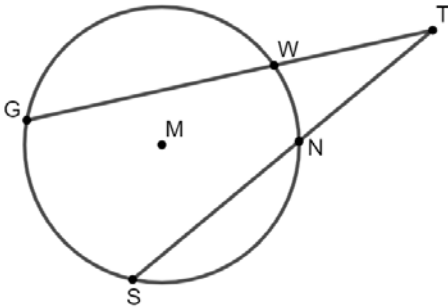
Circle L is shown, where points G , X , A , and H lie on circle L . Secants $\overline{BX}$ and $\overline{BA}$ intersect at point B outside the circle such that $BG = 6$, $GX = 20$, $BH = 4$, and $HA = (6x - 7)$.



4. What is the value of x ?

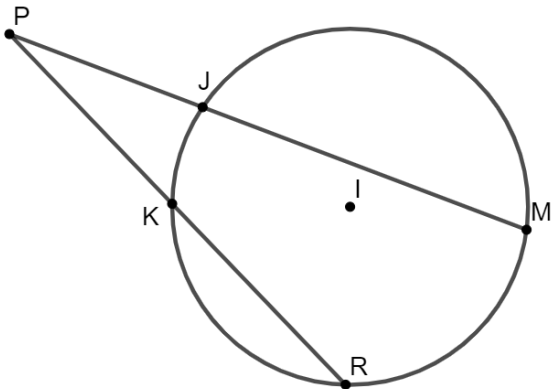
5. Calculate the length of secant $\overline{BA}$.

Circle M with secants $\overline{GT}$ and $\overline{ST}$ is shown. Secants $\overline{GT}$ and $\overline{ST}$ intersect outside the circle at point T so that $GW = 12x$, $WT = 4x$, $SN = 20$, and $NT = 16$.



- 6. Find the value of x .
- 7. What is the length of secant $\overline{GT}$?

Carter and Isabella were solving for x in circle I where points J , M , R , and K are on the circle and secants $\overline{PM}$ and $\overline{PR}$ intersect at point P outside the circle. They know $PJ = 3$, $JM = (4x - 1)$, $PK = 4$, and $KR = 2x$.



Each of their equations are set up to solve for x .

Carter's Work	Isabella's Work
$3(4x - 1) = 4(2x)$	$3(4x + 2) = 4(2x + 4)$

- 8. Whose work is the correct equation needed to solve for x ?
- 9. Use this equation to find the value of x .
- 10. What is the length of secant $\overline{PR}$?